

Jared Schultz

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EDUCATION

University of Minnesota, College of Science and Engineering *Minneapolis, MN*
Master's of Science, Robotics, Minor in Mechanical Engineering *5/2026 – 9/2024*

GPA: 4.00

University of Minnesota, College of Science and Engineering *Minneapolis, MN*
Bachelor of Science, Computer Engineering *9/2020 – 5/2024*

GPA: 3.52

TECHNICAL SKILLS

Robotics: Computer Vision, Machine Learning, AI, Path Planning, Kalman Filtering, Feedback Control, Force Feedback

Programming: C, C++, Java, Python, ROS, Bash, MATLAB, Verilog

Mechanical: SolidWorks, RoboDK, 3d Printing (SLA and FDM), Machine shop trained 100+ hours (mill, lathe, waterjet)

Electrical: DC/AC circuit analysis, Altium PCB Designer, Vivado FPGA

Tools and Platforms: Git, Jira, Confluence, Linux, NumPy, PyTorch, TensorFlow

JOB EXPERIENCE

University of Minnesota and Medtronic *Minneapolis, MN*
Mechatronic Lead for Autonomous Telerobotic Surgical Stroke Robot *6/2025 – 5/2026*

- Built an automated test platform using an ATI Nano17 6-DOF force/torque sensor and a stepper-actuated test rig to characterize prototype performance, accelerate wear, and identify failure modes to promote device longevity.
- Designed, manufactured, and tested iterative robot prototypes, using results to verify and inform design revisions.
- Designed and implemented a motor control architecture enabling precise control of the guidewire propulsion mechanism.

General Dynamics Mission Systems *Bloomington, Minnesota*
Software Engineering Intern *5/2022 – 8/2022*

- Translated the existing test environment from Bash into Python to improve readability and package support.
- Anticipated future requirements and implemented unit and system tests into the active test environment.
- Contributed to professional documentation through Confluence, understood team workflow through Jira, and streamlined version control through GitLab.

PUBLICATIONS

Friction Characterization of a Roller-Drive Mechanism for Robotic Motion Control of Guidewires DMD 2026

Jared Schultz, Pin-Hao Cheng, Matt Rajala, Timothy M. Kowalewski. [PDF] doi.org/10.1115/DMD2026-1086

Towards Remote Thrombectomy with Telerobotically-Driven Guidewires DMD 2026

Pin-Hao Cheng, Ronak Narkhede*, Matt Rajala*, Jared Schultz*, Nathan Harbinson, Samuel Fisher, Nitish Poojari, Sharva Khandagale, Alex Berg, Keara Berlin, Adam Imdieke, Michael Feldkamp, Scott Frushour, Kaustubh Patil, Mark Ashby, William Peine, Sean L. Moen, Andrew Grande, Karthik Desingh, Timothy M. Kowalewski.* [PDF] doi.org/10.1115/DMD2026-1079

* Contributed equally, listed alphabetically by last name

PROJECT EXPERIENCE

Robotics Club projects (Robo Master, NASA Lunabotics, Robot in Three Days) 1/2021 – Present

- Work under a three-day time crunch to design and manufacture a robot to meet competition requirements.
- Created a custom neural stack to identify, track, and predict robot motion in a moving-camera, moving-target environment, achieving 98% classification accuracy with a 3 ms loop time.
- Collaborate with my team to design, manufacture, and validate robotic subsystems to ensure reliable task performance.

Intermediate Robotics Project 9/2024 – 12/2024

- Implemented kinematics, self-motion, path planning, and Jacobian IK for a hyper-redundant planar robot
- Used torque measurements to implement feedback and measure the effects of delay on manipulability
- Developed methods for control, teleoperation, and feedback for UR5 Robot over TCP/IP Ethernet

Robot Vision Final Project 9/2024 – 12/2024

- Designed and built from scratch an autonomous 6 DOF arm able to simulate industrial pick-and-place tasks
- Developed and debugged a custom position-based visual servoing algorithm to allow for dynamic object tracking.

Autonomous Quadrotor for Education 12/2023 – 5/2024

- Assisted in exploratory research, completing the initial steps in creating a new class for autonomous aerial flight.
- Implemented signal processing and path planning algorithms in a pipeline to allow the real-time position of the drone to be used to follow a path autonomously.